

- 5 (a) magnetic flux = BA
 $= 89 \times 10^{-3} \times 5.0 \times 10^{-2} \times 2.4 \times 10^{-2}$
 $= 1.07 \times 10^{-4} \text{ Wb}$ C1
A1 [2]
- (b) (i) e.m.f. = $\Delta\phi / \Delta t$ C1
(for $\Delta\phi = 1.07 \times 10^{-4} \text{ Wb}$), $\Delta t = 2.4 \times 10^{-2} / 1.8 = 1.33 \times 10^{-2} \text{ s}$ C1
e.m.f. = $(1.07 \times 10^{-4}) / (1.33 \times 10^{-2})$
 $= 8.0 \times 10^{-3} \text{ V}$ A1 [3]
- (ii) current = $8.0 \times 10^{-3} / 0.12$ M1
 $\approx 70 \text{ mA}$ A0 [1]
- (c) force on wire = BIL
 $= 89 \times 10^{-3} \times 70 \times 10^{-3} \times 5.0 \times 10^{-2}$ C1
 $\approx 3 \times 10^{-4} \text{ (N)}$ M1
suitable comment e.g. this force is too / very small (to be felt) A1 [3]
- 6 (a) power / heating depends on I^2 M1